Digital Forensic Case Study

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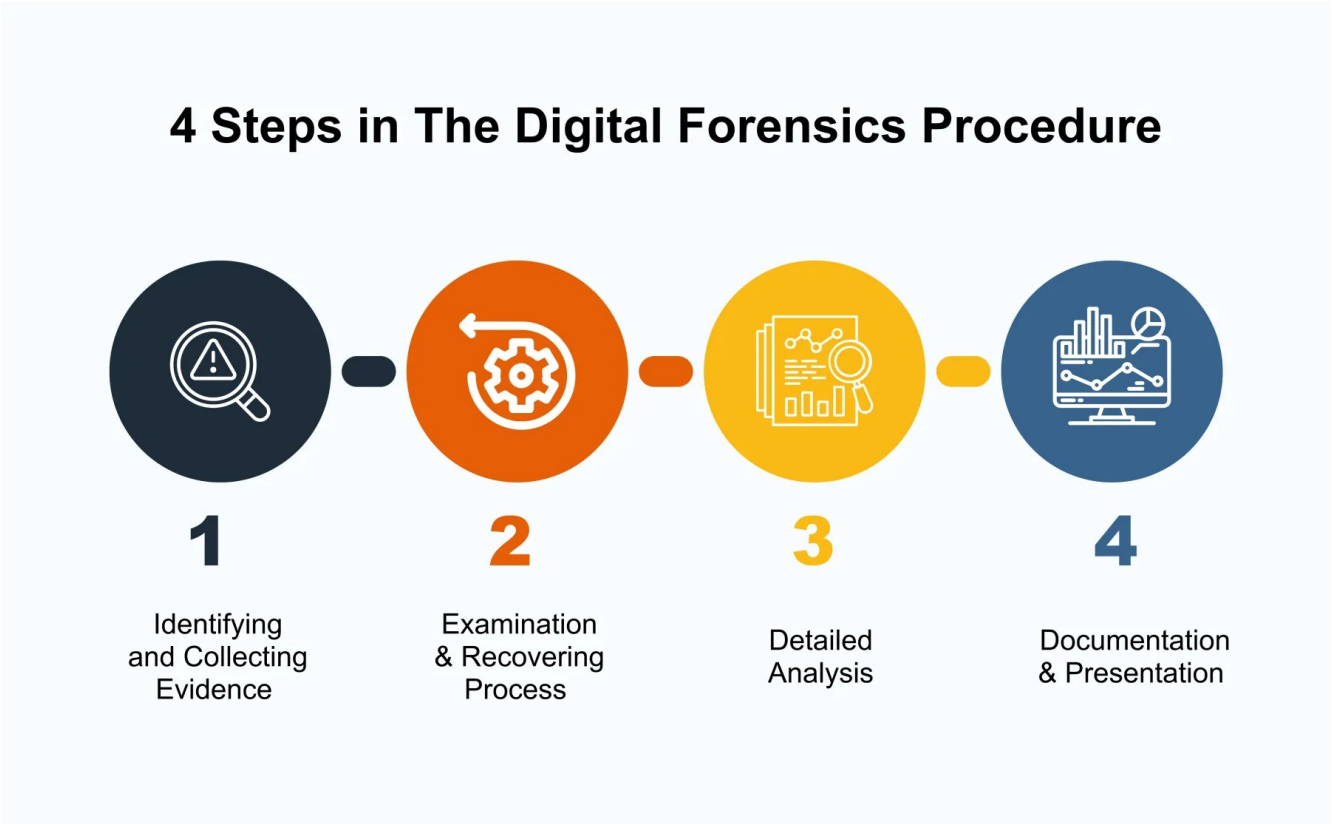

Scenario: Digital Forensics Investigation – Bank Robbery Case

Law enforcement agencies conducted a raid at a suspected bank robber's residence after obtaining a valid search warrant. During the operation, multiple digital devices were seized, including a laptop, mobile phone, external hard drive, and a USB storage device.

All collected evidence was properly documented, preserved using chain of custody procedures, and transferred to a digital forensics laboratory for detailed analysis.

The digital forensics team used industry-standard forensic tools to examine the devices and recovered the following evidence:

- A **digital map of the bank** was found on the suspect's laptop, indicating pre-planning of the robbery.

- A **document containing the bank’s entrance and escape routes** was recovered from the external hard drive.
- Several **media files (photos and videos)** of previous robberies were discovered on the laptop.

Collection:

Find the following details:

1.File Creation

2. Camera model name

```
root@ip-10-114-106-222:~/Rooms/introdigitalforensics# exiftool let
ter-image.jpg
ExifTool Version Number         : 12.76
File Name                       : letter-image.jpg
Directory                       : .
File Size                       : 127 kB
File Modification Date/Time     : 2022:02:23 08:53:33+00:00
File Access Date/Time           : 2026:03:20 03:24:48+00:00
File Inode Change Date/Time     : 2026:02:25 17:59:37+00:00
File Permissions                : -rwxr-xr-x
File Type                       : JPEG
File Type Extension             : jpg
MIME Type                       : image/jpeg
JFIF Version                    : 1.01
Exif Byte Order                 : Little-endian (Intel, II)
Compression                     : JPEG (old-style)
Make                            : Canon
Camera Model Name               : Canon EOS R6
Orientation                     : Horizontal (normal)
```

01

02

3.Exact location (GPS coordinates)

```
Create Date                     : 2022:02:25 13:37:33.42+03:00
Date/Time Original              : 2022:02:25 13:37:33.42+03:00
Modify Date                     : 2022:02:15 17:23:40+01:00
Thumbnail Image                 : (Binary data 4941 bytes, use -b
option to extract)
GPS Latitude                    : 51 deg 30' 51.90" N
GPS Longitude                   : 0 deg 5' 38.73" W
Date/Time Created               : 2022:02:15 17:23:40-17:23
Digital Creation Date/Time      : 2021:11:05 14:06:13+03:00
Circle Of Confusion             : 0.043 mm
Depth Of Field                  : 0.06 m (0.76 - 0.82 m)
Field Of View                   : 54.9 deg
Focal Length                    : 50.0 mm (35 mm equivalent: 34.6
mm)
GPS Position                    : 51 deg 30' 51.90" N, 0 deg 5' 3
.73" W
Light Value                     : 7.9
Lens ID                         : Canon EF 50mm f/1.8 STM
```

03

1. Author Name
2. Creator

```
Title:           Pay Now
Subject:         We Have Gato
Author:          Ann Gree Shepherd
Creator:         Microsoft® Word 2016
Producer:        Microsoft® Word 2016
CreationDate:    Wed Feb 23 09:10:36 2022 UTC
ModDate:         Wed Feb 23 09:10:36 2022 UTC
Custom Metadata: no
Metadata Stream: yes
Tagged:          yes
UserProperties:  no
Suspects:        no
Form:            none
JavaScript:      no
Pages:           1
Encrypted:       no
Page size:       595.44 x 842.04 pts (A4)
Page rot:        0
File size:       71371 bytes
Optimized:       no
PDF version:     1.7
```

Tools Used:

pdfinfo tool
exinfo tool